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(a)	Upthrust is a force on a partially or fully submerged object , acting in the opposite direction to weight
	Equal in magnitude to the weight of the fluid displaced
(b)	$\text{Upthrust} = \rho_{water} Vg$ $= (1000)(0.40)(0.50)(9.81)$ $= 1962 \text{ N}$
	$T_{BE} = \text{weight} - \text{upthrust}$ $= (950)(9.81) - 1962$ $= 7357.5 \text{ N}$
	$= 7360 \text{ N (shown)}$
(c)(i)	Beam is in equilibrium. Taking moments about A, Total clockwise moment = total anticlockwise moment $T_{BE}(3.00\cos 58°) + m_{beam} g \left(\frac{3.00}{2}\cos 58°\right) = T_{CD}(1.24\sin 32°)$ where 1.24 sin 32° = perpendicular distance from CD to O
	$T_{CD} = \frac{(7360)(3.00\cos 58°) + (80.0)(9.81)(1.50\cos 58°)}{1.24\sin 32°}$ $= 18\ 700 \text{ N}$
(c)(ii)	Net force on bar must be zero in all directions for translational equilibrium
	Possible justification includes • *F* must have an upward component to cancel out the net downward force from T_{CD}, T_{BE}, and the weight of the beam. • *F* must have a rightward component to cancel out the leftward force from T_{CD}.
(c)(iii)	Beam is in equilibrium. Taking moments about D, T_{BE} provides clockwise moment.
	Hence, *F* must provide anticlockwise moment, so the direction of *F* is below AB.

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